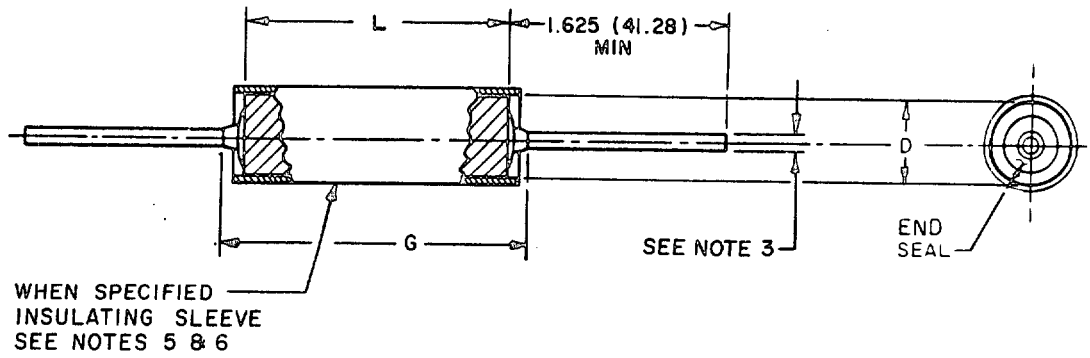
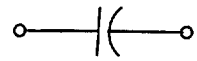
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, SUPERMETALLIZED PLASTIC FILM DIELECTRIC, DIRECT CURRENT, FOR LOW ENERGY, HIGH IMPEDANCE APPLICATIONS, HERMETICALLY SEALED IN METAL CASES, HIGH RELIABILITY, STYLES CHS01, CHS02, AND CHS03

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the capacitors described herein shall consist of this specification and the latest issue of MIL-C-87217.





CIRCUIT DIAGRAM

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. Number 22 AWG wire (.025 (0.64 mm) inches ±.002 (0.05 mm)) for case diameters of .312 (7.92 mm) and less.
   Number 20 AWG wire (.032 (0.81 mm) inches ±.002 (0.05 mm)) for case diameters of .400 (10.16 mm) and .500 (12.70 mm).
   Number 18 AWG wire (.040 (1.02 mm) inches ±.002 (0.05 mm)) for case diameters of .562 (14.27 mm) and over.
4. See table I for additional dimensions.
5. Insulating sleeve shall extend beyond the capacitor body. Insulating sleeve thickness shall not exceed .005 (0.13 mm).
6. Plastic insulating sleeve shall be transparent; marking shall be applied to the capacitor case.
7. Lead length may be a minimum of one inch for use in tape and reel packaging, when specified in the ordering data.

FIGURE 1. Capacitor configuration.

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TABLE I. Electrical characteristics, dimensions, and dash numbers.

CHS01 - 30 volts (dc rating)										
Capacitance value (nom) (in µF)	Dimensions 1/ (in inches with mm in parentheses)			Dash number 2/						
		D	G	Capacitance tolerance value (in %)						
	L ±.030(.76)	+.020(.51) -.010(.25)	(max)	±0.25	±0.5	±1.0	±2.0	±5.0	±10.0
0.001	0.500(12.70)	0.170(4.32)	0.700(17.78)	1001-	1002-	1003-	1004-	1005-	1006-
0.0012	"	"	"	1007-	1008-	1009-	1010-	1011-	1012-
0.0015	"	"	"	1013-	1014-	1015-	1016-	1017-	1018-
0.0018	"	"	"	1019-	1020-	1021-	1022-	1023-	1024-
0.002	"	"	"	1025-	1026-	1027-	1028-	1029-	1030-
0.0022	"	"	"	1031-	1032-	1033-	1034-	1035-	1036-
0.0027	"	"	"	1037-	1038-	1039-	1040-	1041-	1042-
0.0033	"	"	"	1043-	1044-	1045-	1046-	1047-	1048-
0.0039	"	"	"	1049-	1050-	1051-	1052-	1053-	1054-
0.0047	"	"	"	1055-	1056-	1057-	1058-	1059-	1060-
0.005	"	"	"	1061-	1062-	1063-	1064-	1065-	1066-
0.0056	"	"	"	1067-	1068-	1069-	1070-	1071-	1072-
0.0068	"	"	"	1073-	1074-	1075-	1076-	1077-	1078-
0.0082	"	"	"	1079-	1080-	1081-	1082-	1083-	1084-
0.01	"	"	"	1085-	1086-	1087-	1088-	1089-	1090-
0.012	"	"	"	1091-	1092-	1093-	1094-	1095-	1096-
0.015	"	"	"	1097-	1098-	1099-	1100-	1101-	1102-
0.018	"	"	"	1103-	1104-	1105-	1106-	1107-	1108-
0.02	"	"	"	1109-	1110-	1111-	1112-	1113-	1114-
0.022	"	"	"	1115-	1116-	1117-	1118-	1119-	1120-
0.027	0.562(14.27)	"	0.762(19.35)	1121-	1122-	1123-	1124-	1125-	1126-
0.033	"	"	"	1127-	1128-	1129-	1130-	1131-	1132-
0.039	"	"	"	1133-	1134-	1135-	1136-	1137-	1138-
0.047	"	"	"	1139-	1140-	1141-	1142-	1143-	1144-
0.05	"	"	"	1145-	1146-	1147-	1148-	1149-	1150-
0.056	"	"	"	1151-	1152-	1153-	1154-	1155-	1156-
0.068	0.687(17.45)	"	0.887(22.53)	1157-	1158-	1159-	1160-	1161-	1162-
0.082	"	"	"	1163-	1164-	1165-	1166-	1167-	1168-
0.1	"	0.193(4.90)	"	1169-	1170-	1171-	1172-	1173-	1174-
0.12	"	0.193(4.90)	"	1175-	1176-	1177-	1178-	1179-	1180-
0.15	0.562(14.27)	0.235(5.97)	0.762(19.35)	1181-	1182-	1183-	1184-	1185-	1186-
0.18	0.687(17.45)	"	0.887(22.53)	1373-	1374-	1375-	1376-	1377-	1378-
0.20	"	"	0.887(22.53)	1187-	1188-	1189-	1190-	1191-	1192-
0.22	"	"	"	1193-	1194-	1195-	1196-	1197-	1198-
0.27	"	"	"	1199-	1200-	1201-	1202-	1203-	1204-
0.33	"	"	"	1205-	1206-	1207-	1208-	1209-	1210-
0.39	"	"	"	1211-	1212-	1213-	1214-	1215-	1216-
0.47	"	0.312(7.92)	"	1217-	1218-	1219-	1220-	1221-	1222-
0.50	"	"	"	1223-	1224-	1225-	1226-	1227-	1228-
0.56	"	"	"	1229-	1230-	1231-	1232-	1233-	1234-
0.68	"	"	"	1235-	1236-	1237-	1238-	1239-	1240-
0.82	0.813(20.65)	"	1.013(25.73)	1241-	1242-	1243-	1244-	1245-	1246-
1.0	"	"	"	1247-	1248-	1249-	1250-	1251-	1252-
1.2	"	"	"	1253-	1254-	1255-	1256-	1257-	1258-
1.5	"	0.400(10.16)	"	1259-	1260-	1261-	1262-	1263-	1264-
1.8	"	"	"	1265-	1266-	1267-	1268-	1269-	1270-
2.0	"	"	"	1271-	1272-	1273-	1274-	1275-	1276-
2.2	"	"	"	1277-	1278-	1279-	1280-	1281-	1282-
2.7	1.063(27.00)	"	1.263(32.08)	1385-	1386-	1387-	1388-	1389-	1390-
3.0	"	"	"	1289-	1290-	1291-	1292-	1293-	1294-
3.3	"	"	"	1295-	1296-	1297-	1298-	1299-	1300-
3.9	1.375(34.93)	"	1.575(40.01)	1301-	1302-	1303-	1304-	1305-	1306-
4.7	"	0.500(12.70)	"	1313-	1314-	1315-	1316-	1317-	1318-
5.0	"	"	"	1319-	1320-	1321-	1322-	1323-	1324-
5.6	"	"	"	1325-	1326-	1327-	1328-	1329-	1330-
6.8	"	0.562(14.27)	"	1331-	1332-	1333-	1334-	1335-	1336-
8.0	"	"	"	1337-	1338-	1339-	1340-	1341-	1342-
8.2	"	"	"	1343-	1344-	1345-	1346-	1347-	1348-
10.0	"	"	"	1349-	1350-	1351-	1352-	1353-	1354-

See footnotes at end of table.

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TABLE I. Electrical characteristics, dimensions, and dash numbers - Continued.

CHS02 - 50 volts (dc rating)										
Capacitance value (nom) (in µF)	Dimensions 1/ (in inches with mm in parentheses)			Dash number 2/						
		D	G	Capacitance tolerance value (in %)						
	L ±.030(.76)	+.020(.51) -.010(.25)	(max)	±0.25	±0.5	±1.0	±2.0	±5.0	±10.0
0.001	0.500(12.70)	0.170(4.32)	0.700(17.78)	2001-	2002-	2003-	2004-	2005-	2006-
0.0012	"	"	"	2007-	2008-	2009-	2010-	2011-	2012-
0.0015	"	"	"	2013-	2014-	2015-	2016-	2017-	2018-
0.0018	"	"	"	2019-	2020-	2021-	2022-	2023-	2024-
0.002	"	"	"	2025-	2026-	2027-	2028-	2029-	2030-
0.0022	"	"	"	2031-	2032-	2033-	2034-	2035-	2036-
0.0027	"	"	"	2037-	2038-	2039-	2040-	2041-	2042-
0.0033	"	"	"	2043-	2044-	2045-	2046-	2047-	2048-
0.0039	"	"	"	2049-	2050-	2051-	2052-	2053-	2054-
0.0047	"	"	"	2055-	2056-	2057-	2058-	2059-	2060-
0.005	"	"	"	2061-	2062-	2063-	2064-	2065-	2066-
0.0056	"	"	"	2067-	2068-	2069-	2070-	2071-	2072-
0.0068	"	"	"	2073-	2074-	2075-	2076-	2077-	2078-
0.0082	0.562(14.27)	"	0.762(19.35)	2079-	2080-	2081-	2082-	2083-	2084-
0.01	"	"	"	2085-	2086-	2087-	2088-	2089-	2090-
0.012	"	"	"	2091-	2092-	2093-	2094-	2095-	2096-
0.015	"	"	"	2097-	2098-	2099-	2100-	2101-	2102-
0.018	"	0.193(4.90)	"	2103-	2104-	2105-	2106-	2107-	2108-
0.02	"	"	"	2109-	2110-	2111-	2112-	2113-	2114-
0.022		"		2115-	2116-	2117-	2118-	2119-	2120-
0.027	"	"	"	2121-	2122-	2123-	2124-	2125-	2126-
0.033	"	"	"	2127-	2128-	2129-	2130-	2131-	2132-
0.039	0.687(17.45)	0.170(4.32)	0.887(22.53)	2133-	2134-	2135-	2136-	2137-	2138-
0.047	"	"	"	2139-	2140-	2141-	2142-	2143-	2144-
0.05	"	"	"	2145-	2146-	2147-	2148-	2149-	2150-
0.056	"	0.193(4.90)	"	2151-	2152-	2153-	2154-	2155-	2156-
0.068	"	"	"	2157-	2158-	2159-	2160-	2161-	2162-
0.082	0.813(20.65)	"	1.013(25.73)	2163-	2164-	2165-	2166-	2167-	2168-
0.1	0.813(20.65)	"	1.013(25.73)	2169-	2170-	2171-	2172-	2173-	2174-
0.12	0.687(17.45)	0.235(5.97)	0.887(22.53)	2175-	2176-	2177-	2178-	2179-	2180-
0.15	0.687(17.45)	"	0.887(22.53)	2181-	2182-	2183-	2184-	2185-	2186-
0.18	0.813(20.65)	"	1.013(25.73)	2187-	2188-	2189-	2190-	2191-	2192-
0.20	"	"	"	2193-	2194-	2195-	2196-	2197-	2198-
0.22	"	"	"	2199-	2200-	2201-	2202-	2203-	2204-
0.27	0.687(17.45)	0.312(7.92)	0.887(22.53)	2205-	2206-	2207-	2208-	2209-	2210-
0.33	0.687(17.45)	"	0.887(22.53)	2211-	2212-	2213-	2214-	2215-	2216-
0.39	0.813(20.65)	"	1.013(25.73)	2217-	2218-	2219-	2220-	2221-	2222-
0.47	"	"	"	2223-	2224-	2225-	2226-	2227-	2228-
0.50	"	"	"	2229-	2230-	2231-	2232-	2233-	2234-
0.56	"	0.400(10.16)	"	2235-	2236-	2237-	2238-	2239-	2240-
0.68	"	"	"	2241-	2242-	2243-	2244-	2245-	2246-
0.82	1.063(27.00)	"	1.263(32.08)	2247-	2248-	2249-	2250-	2251-	2252-
1.0	"	"	"	2253-	2254-	2255-	2256-	2257-	2258-
1.2	"	"	"	2259-	2260-	2261-	2262-	2263-	2264-
1.5	"	"	"	2265-	2266-	2267-	2268-	2269-	2270-
1.8	"	"	"	2271-	2272-	2273-	2274-	2275-	2276-
2.0	1.125(28.58)	0.500(12.70)	1.325(33.66)	2277-	2278-	2279-	2280-	2281-	2282-
2.2	1.125(28.58)	"	1.325(33.66)	2283-	2284-	2285-	2286-	2287-	2288-
2.7	1.375(34.93)	"	1.575(40.01)	2289-	2290-	2291-	2292-	2293-	2294-
3.0	"	"	"	2295-	2296-	2297-	2298-	2299-	2300-
3.3	"	"	"	2301-	2302-	2303-	2304-	2305-	2306-
3.9	"	0.562(14.27)	"	2307-	2308-	2309-	2310-	2311-	2312-
4.7	"	0.670(17.02)	"	2319-	2320-	2321-	2322-	2323-	2324-
5.0	"	"	"	2325-	2326-	2327-	2328-	2329-	2330-
5.6	"	"	"	2331-	2332-	2333-	2334-	2335-	2336-
6.8	1.875(47.63)	"	2.075(52.71)	2337-	2338-	2339-	2340-	2341-	2342-
8.0	"	"	"	2343-	2344-	2345-	2346-	2347-	2348-
8.2	"	"	"	2349-	2350-	2351-	2352-	2353-	2354-
10.0	"	"	"	2355-	2356-	2357-	2358-	2359-	2360-

See footnotes at end of table.

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TABLE I. Electrical characteristics, dimensions, and dash numbers - Continued.

CHS03 - 100 volts (dc rating)										
Capacitance value (nom) (in µF)	Dimensions 1/ (in inches with mm in parentheses)			Dash number 2/						
		D	G	Capacitance tolerance value (in %)						
	L ±.030(.76)	+.020(.51) -.010(.25)	(max)	±0.25	±0.5	±1.0	±2.0	±5.0	±10.0
0.001	0.500(12.70)	0.170(4.32)	0.700(17.78)	3001-	3002-	3003-	3004-	3005-	3006-
0.0012	"	"	"	3007-	3008-	3009-	3010-	3011-	3012-
0.0015	"	"	"	3013-	3014-	3015-	3016-	3017-	3018-
0.0018	"	"	"	3019-	3020-	3021-	3022-	3023-	3024-
0.002	"	"	"	3025-	3026-	3027-	3028-	3029-	3030-
0.0022	"	"	"	3031-	3032-	3033-	3034-	3035-	3036-
0.0027	"	"	"	3037-	3038-	3039-	3040-	3041-	3042-
0.0033	"	"	"	3043-	3044-	3045-	3046-	3047-	3048-
0.0039	"	"	"	3049-	3050-	3051-	3052-	3053-	3054-
0.0047	"	"	"	3055-	3056-	3057-	3058-	3059-	3060-
0.005	"	"	"	3061-	3062-	3063-	3064-	3065-	3066-
0.0056	"	"	"	3067-	3068-	3069-	3070-	3071-	3072-
0.0068	0.562(14.27)	"	0.762(19.35)	3073-	3074-	3075-	3076-	3077-	3078-
0.0082	0.562(14.27)	"	0.762(19.35)	3079-	3080-	3081-	3082-	3083-	3084-
0.01	0.687(17.45)	"	0.887(22.53)	3085-	3086-	3087-	3088-	3089-	3090-
0.012	"	"	"	3091-	3092-	3093-	3094-	3095-	3096-
0.015	"	"	"	3097-	3098-	3099-	3100-	3101-	3102-
0.018	"	0.193(4.90)	"	3103-	3104-	3105-	3106-	3107-	3108-
0.02	"	"	"	3109-	3110-	3111-	3112-	3113-	3114-
0.022		"		3115-	3116-	3117-	3118-	3119-	3120-
0.027	"	"	"	3121-	3122-	3123-	3124-	3125-	3126-
0.033	"	"	"	3127-	3128-	3129-	3130-	3131-	3132-
0.039	"	0.235(5.97)	"	3133-	3134-	3135-	3136-	3137-	3138-
0.047	"	"	"	3139-	3140-	3141-	3142-	3143-	3144-
0.050	"	"	"	3145-	3146-	3147-	3148-	3149-	3150-
0.056	"	"	"	3151-	3152-	3153-	3154-	3155-	3156-
0.068	0.813(20.65)	"	1.013(25.73)	3157-	3158-	3159-	3160-	3161-	3162-
0.082	0.687(17.45)	0.312(7.92)	0.887(22.53)	3163-	3164-	3165-	3166-	3167-	3168-
0.10	"	"	"	3169-	3170-	3171-	3172-	3173-	3174-
0.12	"	"	"	3175-	3176-	3177-	3178-	3179-	3180-
0.15	0.813(20.65)	"	1.013(25.73)	3181-	3182-	3183-	3184-	3185-	3186-
0.18	"	"	"	3187-	3188-	3189-	3190-	3191-	3192-
0.20	"	"	"	3193-	3194-	3195-	3196-	3197-	3198-
0.22	"	"	"	3199-	3200-	3201-	3202-	3203-	3204-
0.27	1.063(27.00)	"	1.263(32.08)	3205-	3206-	3207-	3208-	3209-	3210-
0.33	"	"	"	3211-	3212-	3213-	3214-	3215-	3216-
0.39	"	0.400(10.16)	"	3217-	3218-	3219-	3220-	3221-	3222-
0.47	"	"	"	3223-	3224-	3225-	3226-	3227-	3228-
0.5	"	"	"	3229-	3230-	3231-	3232-	3233-	3234-
0.56	"	"	"	3235-	3236-	3237-	3238-	3239-	3240-
0.68	1.125(28.58)	0.500(12.70)	1.325(33.66)	3241-	3242-	3243-	3244-	3245-	3246-
0.82	"	0.500(12.70)	"	3247-	3248-	3249-	3250-	3251-	3252-
1.00	"	0.562(14.27)	"	3253-	3254-	3255-	3256-	3257-	3258-
1.20	"	"	"	3259-	3260-	3261-	3262-	3263-	3264-
1.50	1.375(34.93)	"	1.575(40.01)	3265-	3266-	3267-	3268-	3269-	3270-
2.0	"	0.670(17.02)	"	3271-	3272-	3273-	3274-	3275-	3276-
2.2	"	"	"	3277-	3278-	3279-	3280-	3281-	3282-
2.7	1.875(47.63)	"	2.075(52.71)	3283-	3284-	3285-	3286-	3287-	3288-
3.0	"	"	"	3289-	3290-	3291-	3292-	3293-	3294-
3.3	"	"	"	3295-	3296-	3297-	3298-	3299-	3300-
3.9	"	0.750(19.05)	"	3301-	3302-	3303-	3304-	3305-	3306-
4.7	"	"	"	3313-	3314-	3315-	3316-	3317-	3318-
5.0	"	"	"	3319-	3320-	3321-	3322-	3323-	3324-
5.6	"	"	"	3325-	3326-	3327-	3328-	3329-	3330-
6.8	2.375(60.53)	1.000(25.40)	2.575(65.41)	3331-	3332-	3333-	3334-	3335-	3336-
8.0	"	"	"	3337-	3338-	3339-	3340-	3341-	3342-
8.2	"	"	"	3343-	3344-	3345-	3346-	3347-	3348-
10.0	"	"	"	3349-	3350-	3351-	3352-	3353-	3354-

1/ L and D are bare case dimensions (see figure 1).
2/ Single letter A = sleeved or B = unsleeved.

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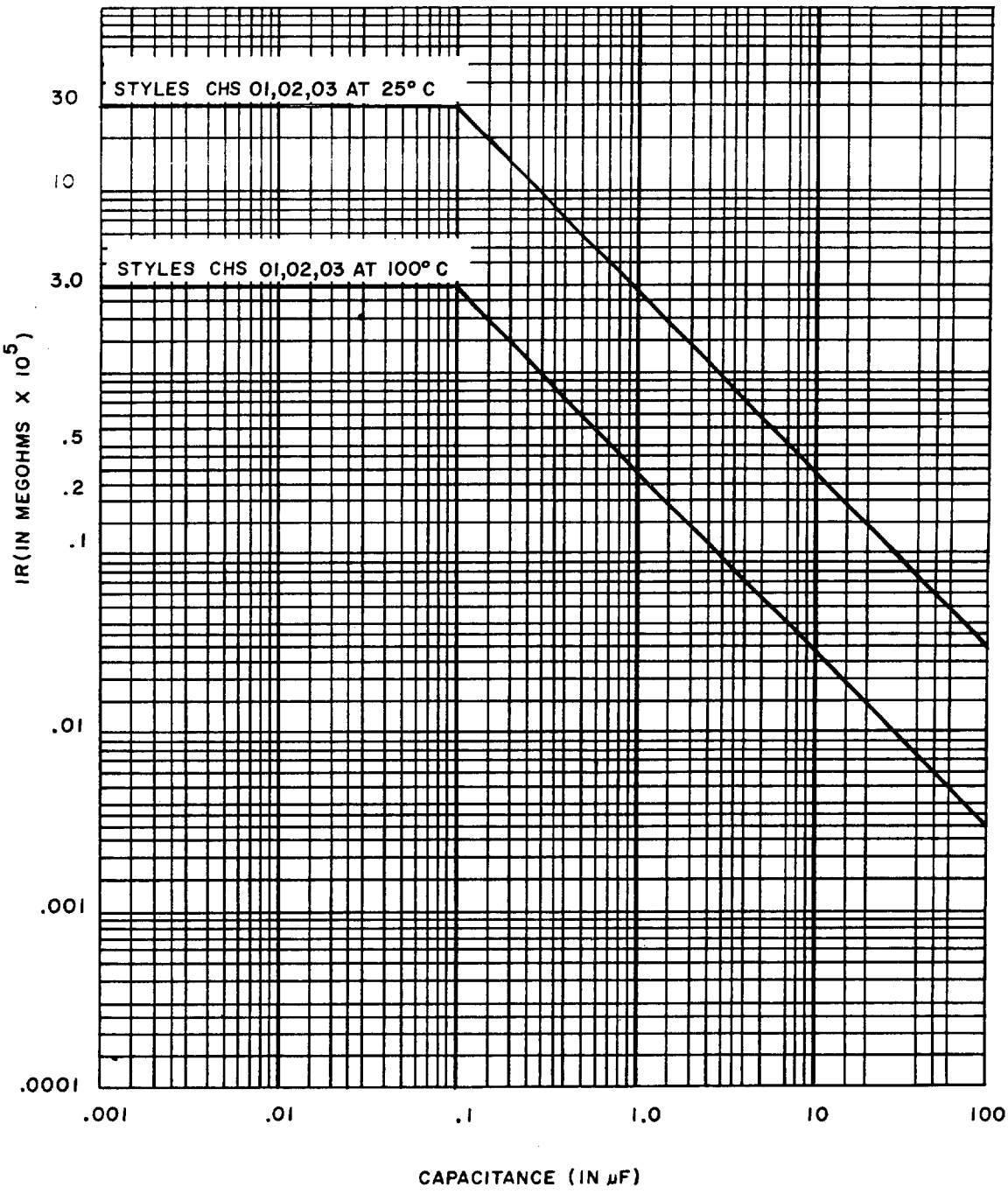


FIGURE 2. Insulation resistance vs. capacitance value.

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REQUIREMENTS:

Parts supplied to this specification shall be qualified to "S" failure rate level and shall have passed subgroups 1 and 2 in group A inspection of MIL-C-83421. All parts shall be permanently serialized on the case prior to group A inspection of MIL-C-87217.

Design and construction:

Dimensions and configuration: See figure 1 and table I.

Case material: Nonmagnetic.

Leads

.500 inch and smaller case diameters: Solder-coated copper clad steel. Type N2 of MIL-STD-1276 may be furnished when specified in the contract or purchase order (see MIL-C-87217 ordering data).

.562 inch and larger case diameters: Type C of MIL-STD-1276.

Capacitance (Cap.) value: See table I.

Cap. tolerance: See table I.

Rated voltage: See tables I and II.

TABLE II. DC ratings.

Styles	Volts, dc
CHS01	30
CHS02	50
CHS03	100

Operating temperature range: -55°C to +100°C.

Thermal shock: In accordance with MIL-C-87217 with the following exception:

20 cycles, -55°C to 100°C, nonenergized with a 5-minute maximum wait between chambers.

DC burn-in: In accordance with MIL-C-87217 with the following exception:

160 hours minimum at +100°C with 1.5 ±0.5 V dc and 1.0 through 10.0 megohms resistor in series with each capacitor.

High impedance temperature and voltage ramp test: In accordance with MIL-C-87217.

Insulation resistance (IR): Method 302 of MIL-STD-202. The following details shall apply:

a. Test potential: Rated voltage.

b. Points of measurement:

Terminal to terminal: Between terminals at the applicable specified temperatures, and at 25°C ±3°C.

c. Charge time 5 minutes maximum; however, for capacitance values greater than 1.0µF, an additional 1 minute per µF is permitted.

Capacitance: Method 305 of MIL-STD-202. The following details shall apply:

a. Test frequency: 1,000 ±100 hertz (Hz).

b. Limit of accuracy: ±0.05 percent.

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Dissipation factor (DF): The dissipation factor shall be measured at 1000 ±100 Hz (for capacitors having a nominal capacitance of 1 microfarad or less) or 100 ±10 Hz (for capacitors having a nominal capacitance greater than 1 microfarad).

Seal: Method 112 of MIL-STD-202, test condition C, procedure IIIa, followed by test condition A.

Leakage-rate sensitivity: 10^{-6} atm cm^3/s.

Radiographic inspection: In accordance with MIL-C-87217.

High impedance dc life: The capacitors shall be tested 250 hours at 100°C with 1.5 ±0.5 volts dc and 1 through 10 megohms resistor in series with each capacitor so that the energy from source or across the capacitor is 500 microjoules, maximum. The following details shall apply.

a. Insulation resistance:

Terminal to terminal: Shall meet the initial requirements.

b. Capacitance: Shall meet the initial requirements

c. Dissipation factor: Shall meet the initial requirements

Marking: Applicable.

Part number: Consists of the basic number of this specification sheet with a dash number coded as shown in the following:

M87217/01-

1	123	A

Single digit designating style (i.e., 1 - CHS01; 2 = CHS02; and 3 = CHS03).

Nonsignificant dash number from table I.

Single letter A = sleeved or B = unsleeved.

Packaging: Capacitors will be furnished in tape and reel packaging when so specified in the ordering data.

Custodians:
Army - ER
Navy - EC
Air Force - 11
NASA - NA

Review activities:
Army - AR, MI
Air Force - 17, 19, 85
DLA - ES

User activities:
Navy - AS, CG, MC, OS, SH

Agent:
DLA - ES

Preparing activity:
Air Force - 11

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